INCH-POUND
MIL-STD-25C
6 January 2026
SUPERSEDING
MIL-STD-25B
10 May 1984

DEPARTMENT OF DEFENSE STANDARD PRACTICE

SHIP STRUCTURAL SYMBOLS FOR USE ON SHIP DRAWINGS

AMSC N/A AREA DRPR

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This standard was developed to provide guidance for assuring uniformity in the preparation of ship structural drawings. The drawings to which it applies include those prepared by the Naval Sea Systems Command and by private contractors, for providing contract requirements or guidance, and for describing construction details.

3. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, D.C. 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This standard applies to ship structural drawings for steel and aluminum ships and supplements ASME Y14.24, ASME Y14.34, ASME Y14.35, ASME Y14.38, and ASME Y14.100. Where a conflict exists between these documents, this standard takes precedence.

1.2 Applicability. Two distinct categories of ship drawings as defined in MIL-STD-31000 are covered by this document as follows:

a. Structural contract drawings and contract guidance drawings are Naval Sea Systems Command drawings used to define or illustrate Navy requirements for a ship structural design as outlined in the ship specifications.

b. Structural ship construction drawings are the detailed drawings from which the individual parts of the ship are fabricated and erected.

For the former, it is generally sufficient to outline the extent of each change in weight of plating and indicate the general location and sizes of component parts of the structure. Ship construction drawings, on the other hand, give a precise and complete description of what is required. This includes exact boundaries of each plate, sizes and location of stiffening members, and joints and attachments.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.38 - Abbreviations and Acronyms for Use in Product Definition and Related Documents

(Copies of this document are available online at www.asme.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 - Standard Symbols for Welding, Brazing, and Nondestructive Examination

(Copies of this document are available online at www.aws.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Standard abbreviations. Frequently used standard abbreviations used in structural drawings are shown in table I. Additional standard abbreviations are presented with the discussions addressing their subject matter elsewhere in this document. ASME Y14.38 provides a complete list of accepted standard abbreviations.

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TABLE I. Frequently used standard abbreviations.

Term	Abbreviation	Term	Abbreviation
above	ABV	intercostal	INTCL
above baseline	ABL	in way of	IWO
air tight	AT	knuckle	KNU
baseline	BL	level	LVL
below	BLW	lightened	LTD
bracket	BRKT	load waterline	LWL
breasthook	BH	longitudinal	LONG. 1/
bulkhead	BHD	longitudinal bulkhead	LBHD
center vertical keel	CVK	nontight	NT
centered	CTR	off centerline	OCL
centerline	CL	plate	PL
cut from	CF	platform	PLATF
deck	DK	reference	REF
design waterline	DWL	schedule	SCHED
detail	DT	similar	SIM
ditto	DO	stanchion	STAN
double extra strong (pipe)	XX STR	standard	STD
extra strong (pipe)	XSTR	station	STA
face plate	FC PL	stiffener	STIF
faying flange	FAY FLG	stringer	STGR
flange	FLG	tank top	TT
flange plate	FLG PL	transverse	TRANSV
flat bar	FB	transverse bulkhead	TBHD
floor	FL	trunk	TRK
frame	FR	typical	TYP
fume tight	FT	unless otherwise noted	UON/UOS
gauge	GA	watertight	WT
innerbottom	IB		

NOTE:

1/ Periods are used only when the abbreviation spells another word.

4. GENERAL REQUIREMENTS

Not applicable.

5. DETAILED REQUIREMENTS

5.1 Delineation. Conventional delineation used on ship structural drawings is illustrated on figure 1.

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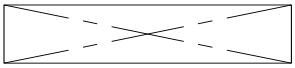
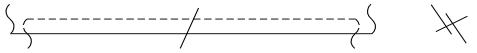
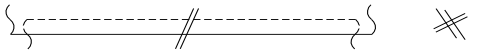
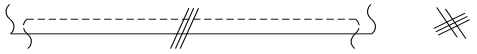
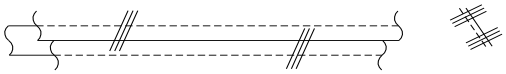
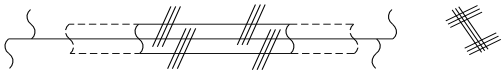
SYMBOL	INTERPRETATION
	DIMENSION LINES, PLATE EDGE AND STIFFENERS, NEAR SIDE
	BULKHEADS, DECKS, OR OTHER PLATING IN SECTION
	PLATE EDGES, FLOORS, GIRDERS, BEAMS, STIFFENERS, ETC., FAR SIDE OR BELOW
	DECKS AND BULKHEADS, FAR SIDE OR BELOW
	CENTERLINE
	OPENING OR SPECIFIC AREA OR BOUNDARY
	WELDED SEAM OR BUTT
	SINGLE FASTENED LAP
	DOUBLE FASTENED LAP
	TRIPLE FASTENED LAP
	FASTENED SINGLE BUTT OR SEAM STRAP
	FASTENED DOUBLE BUTT OR SEAM STRAP

FIGURE 1. Conventional delineation.

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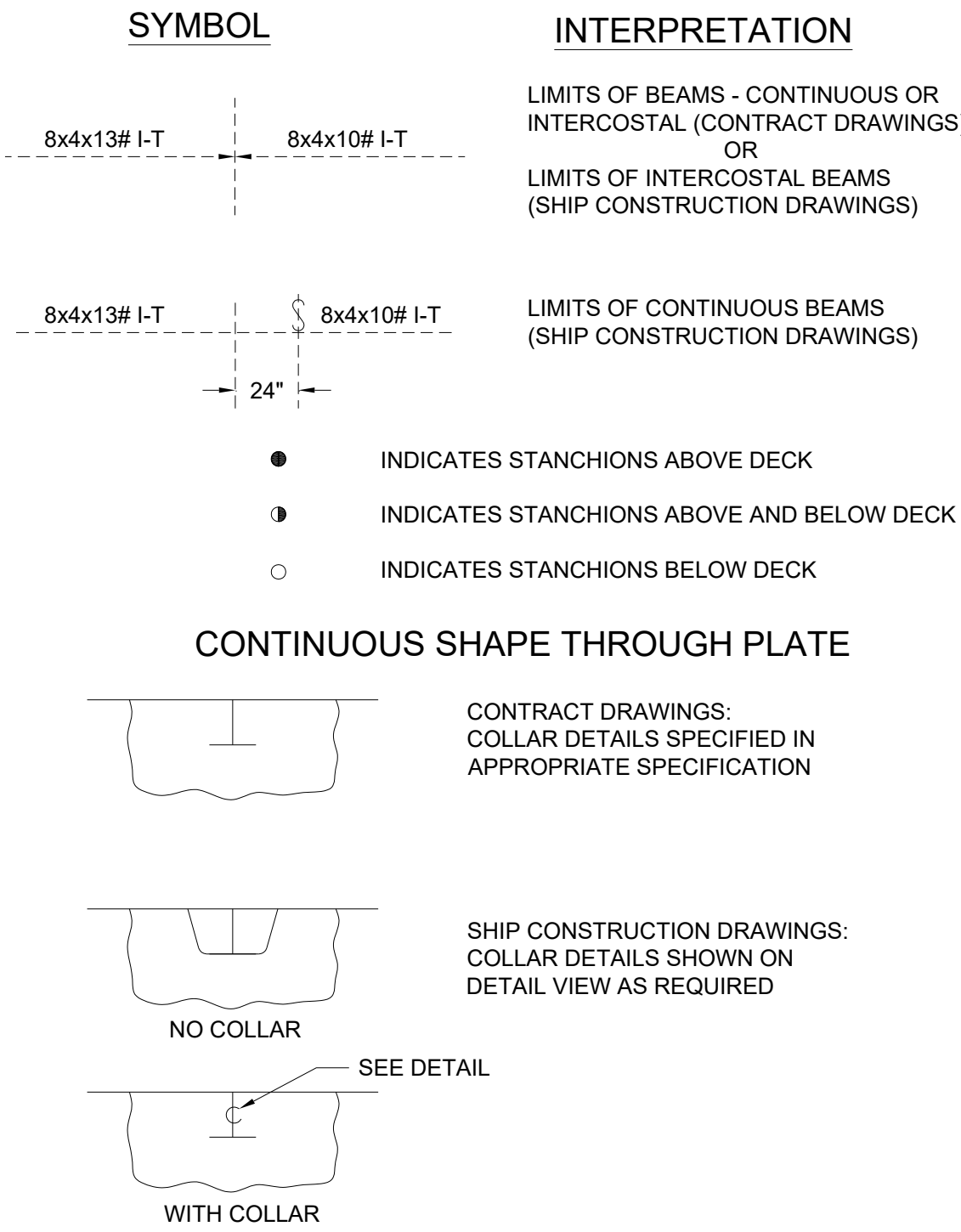


FIGURE 1. Conventional delineation – Continued.

5.2 Designating material. Material shall be indicated by symbols, as listed in table II, following the size designations of the members. However, where the structure is predominantly of one material, this may be indicated by a general note, with exceptions labeled in the normal manner.

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TABLE II. Material symbols.

Material	Symbols
Ordinary strength steel	OSS [1/], OS
High-strength low alloy steel	HSLA-65, HSLA-80, HSLA-100, HSLA-115 [2/]
Higher strength steel	HSS [1/], HS
High yield steel	HY-80, HY-100, HY-130 [2/]
Medium steel	MS [1/]
High tensile steel	HTS [1/]
Special treatment steel	STS [1/]
Corrosion resisting steel	CRES (e.g., 316, 304, 2205, etc.) [3/]
Aluminum alloy	AL (e.g., 5086 H111, 5456 H116, etc.) [3/]

NOTES:

[1/] Obsolete, listed for reference only.

[2/] Numbers refer to the yield strength.

[3/] Numbers refer to the specific alloy.

5.3 Designating size.

5.3.1 Plate thickness. Plate thicknesses, as shown in table III, are designated on drawings. The plate symbol, PL, is omitted for simplicity when the size is labeled directly on the member. For example, a 15.3 pound per square foot ordinary strength steel plate would be labeled 15.3# OSS or where the material is covered by the general notes, 15.3#. If reference is made to a plate and the label is not directly written on the member, the PL symbol should be used. For example, “ALL WEBS SHALL BE 15.3# PL.”

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TABLE III. Plate thickness.

Material	Designation	Examples
Medium steel [1/]	By weight in pounds per square foot of area	20.4# MS
Ordinary strength steel	By weight in pounds per square foot of area	20.4# OS
High-strength low-alloy steel	By weight in pounds per square foot of area	30.6# HSLA-80
Higher strength steel	By weight in pounds per square foot of area	15.3# HS
High tensile steel [1/]	By weight in pounds per square foot of area	15.3# HTS
High yield steel for noncritical applications	By weight in pounds per square foot of area	30.6# HY-80
Corrosion resisting steel	By weight in pounds per square foot of area	7.65# CRES 316
Special treatment steel [1/]	By weight in pounds per square foot of area, rounded to nearest integer	50# STS
Aluminum alloy	By thickness in inches and decimals of an inch	0.75 AL 5456 H116
High yield steel for critical applications [2/]	By thickness in inches and decimals of an inch	1.92 HY-80
Sheet steel	By thickness in inches and decimals of an inch	0.0625 (16 GA)
Other materials less than ⅛-inch thickness	By thickness in inches and decimals of an inch	0.0625 (16 GA)

NOTES:

[1/] Obsolete and listed here for reference only.

[2/] Plate thicknesses are specified in inches and decimals for critical applications that require tighter tolerances on plate thickness or ultrasonic inspection for thickness and soundness.

5.3.2 Standard shapes.

5.3.2.1 Standard shape notation and symbols. The notation and symbols representing rolled or extruded shapes are as shown in table IV.

TABLE IV. Standard shape notation and symbols.

Shape	Notation			Symbol
	First number	Second number	Third number	
I-beams (standard, wide flange sections, light beams, junior beams, and joists)	Web nominal depth in inches and fractions	Flange width in inches and fractions	Linear density in pounds (and decimals) per foot	I, W
Channels (standard, junior, and ship channels)	Web nominal depth in inches and fractions	Flange width in inches and fractions	Linear density in pounds (and decimals) per foot	C
Angles (equal and unequal legs)	Longer leg width in inches and fractions	Shorter leg width in inches and fractions	Thickness in inches and fractions	L
Tees	Web nominal depth in inches and fractions	Flange width in inches and fractions	Linear density in pounds (and decimals) per foot	T, W-T

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TABLE IV. Standard shape notation and symbols – Continued.

Shape	Notation			Symbol
	First number	Second number	Third number	
Flat bars (up to and including 8-inch width; sections exceeding 8 inches are referred to as plates)	Width in inches and fractions	Thickness in inches and fractions	—	FB
Half rounds	Outside diameter in inches and fractions	—	—	HR
Tubing	Outside diameter in inches and diameters	Wall thickness in decimal inches	—	TUBE
Pipe	Nominal diameter in inches and SCHED, STD or strength	—	—	PIPE
Bulb angles	Longer leg width in inches and fractions	Shorter leg width in inches and fractions	Linear density in pounds (and decimals) per foot	BA
Zee bars	Depth in inches and fractions	Leg length in inches and fractions	Linear density in pounds (and decimals) per foot	Z

5.3.2.2 Miscellaneous columns and beams. A few sizes of “miscellaneous columns and beams” have the same nominal depth, flange width, and weight per foot as certain “wide flange sections”, the variation being in the thickness of web and flange. When using these sections, the symbol “I” should not be used and they should be designated either as a “miscellaneous column” or “miscellaneous beam”, whichever applies.

5.3.2.3 Symbols for designating shapes. Application of symbols for designating shapes are given in the following examples:

8 × 5-1/2 × 21#I

8 × 5-1/4 × 20#MISC BEAM

13 × 4 × 40#C

6 × 6 × 1/2L

5 × 3-1/2 × 1/2L

6 × 4 × 7#T

6 × ¼FB

3 HR

6 × 0.25TUBE

6 STD PIPE

6 XSTR PIPE

6 SCHED #PIPE

9 × 3-1/2 × 21.6#BA

4 × 3 × 8.2#Z

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5.3.2.4 Symbol designations. In all designations specified in table IV and 5.3.2.3, inch marks are omitted but symbol "#" is used for clarity to indicate the weight in pounds per linear foot. Nominal sizes are used throughout.

5.3.2.5 Structural tubing. Steel and aluminum structural tubing should not be referred to as pipe since the material specification for pipe does not specify structural physical properties. Tubing is designated by outside diameter and wall thickness. Structural steel tubing is sometimes more readily available in nominal pipe sizes in which case it should be designated by nominal diameter and schedule, standard, or strength.

5.3.3 Standard shapes cut from rolled sections.

5.3.3.1 Shapes with unchanged section depth. Special shapes can be made from rolled I-beams, wide flange shapes, or channels by cutting off either one or two sides of a flange. This leaves the depth of the section unchanged. Shape dimensional notation is as shown in table IV. The most common applications and symbols representing each are listed in table V.

TABLE V. Symbols for shapes with unchanged section depth.

Application	Symbol
I-beam, or wide flange shapes cut to a T (two sides of one flange removed)	I-T
I-beam, or wide flange shapes cut to a J (one side of one flange removed)	I-J
Channel cut to an angle	C-L

Examples:

12 × 4 × 19#I-T

10 × 8 × 33#I-J

13 × 4 × 31.8#C-L

5.3.3.2 Shapes with half nominal depth. Many tees are manufactured from I-beams by cutting through the middle of the web to make two tees. Such members are indicated as tees of half the nominal depth and half the weight of the parent beam.

Example:

6 × 4 × 7#T (This is a T cut from a 12 × 4 × 14#I)

5.3.3.3 Serrated shapes. Serrated angles and tees can be made from channels and I-beams, respectively, by making a serrated cut lengthwise through the web. Since no universal standard for the configuration has been established, details of the cuts shall be shown on the drawing. The symbols listed in table VI shall be used to indicate serrated members.

TABLE VI. Symbols for serrated shapes.

Application	Symbol
Tee-beam serrated from an I	I-SERR(*)T
Angle serrated from a channel	C-SERR(*)L
NOTE: Indicate the finish depth of T or L in place of *.	

Examples:

10 × 4 × 17# I-SERR 7"T

12 × 4 × 35# C-SERR 8"L

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5.3.4 Special tees and angles.

5.3.4.1 I-beam faying flange. An I-beam or channel can be modified by trimming off a part of one flange to leave the remainder of the faying flange intact. As an example, for a deck longitudinal or bulkhead stiffener requiring a 4-inch faying flange, the following nomenclature shall be used:

12 × 6-1/2 × 26#I-T (4 FAY FLG)

5.3.4.2 Channel faying flange. For channels requiring a faying flange, the following nomenclature shall be used:

13 × 4 × 35#C-C (2-1/2 FAY FLG)

5.3.4.3 Shapes cut from larger shapes. To modify a member where the above convention does not apply the symbol CF (cut from) can be used.

Examples:

16 × 4L CF 18 × 4 × 42.7#C

4-1/2 × 1-1/4L CF 5 × 3 × 1/4L

6 × 2L CF 6 × 3-1/2 × 1/2L

5.3.5 Built-up tees.

5.3.5.1 Built-up tee designations. Tee sections may be built up by welding plates and flat bars together. Built-up tees are indicated by designating the overall depth of tee section, by width of flange, followed by the weight per square foot of the web and the weight per square foot of the flange. For high-yield strength steel tees used in critical application or aluminum alloy, the flange and web thickness are expressed in decimals. A tee with an overall depth of 18 inches giving a 20.4# web and a 30.6# flange 7 inches wide is represented as follows:

18 × 7 × 20.4#/30.6#T (steel)

18 × 0.5 WEB with 7 × 0.75 FC PL CTR (steel)

18 × 7 × 0.50/0.75 T (aluminum alloy)

5.3.5.2 Designation for critical applications. In critical applications, tee sections may be built up by welded plates or flat bars. A tee with an overall depth of 18 inches having a ½-inch thick web and ¾-inch thick flange with a 7-inch flange width is represented as follows:

18 × 7 × 0.50/0.75 T (critical application)

5.3.6 Flanged plates used as structural members. The term “flange plate” refers to a plate transformed into an angle stiffener using a break point press.

5.3.6.1 Flange plate denotation. Flanged plates used as structural members are indicated by designating the depth of member by width of flange by weight per square foot of plate.

Examples:

18 × 4 × 15.3# FLG PL

18 × 7-1/2 × 20.4# FLG PL

5.4 Shell strakes. The term “shell strake” is applied to a continuous row of shell plating that may be connected to adjacent rows of plating by welding or mechanical fasteners.

5.4.1 Shell strake denotation. On ship construction drawings, in the portion of the ship where these strakes run fore and aft, they are identified by a letter. The letter “A” identifies the first plate adjacent to the flat keel, continued by “B”, “C”, “D”, etc., to the upper part of the shell. The letter “I” is not used, since it may be confused with the numeral one.

5.4.2 Shell strake identification on contract drawings. Contract drawings and contract guidance drawings show only changes in plate thickness and therefore the lettering for individual strakes is not used.

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5.5 Shell and deck longitudinals. The name "longitudinal" is a generic term applying to all stiffening members running in a fore and aft direction. The Navy has adopted the policy of avoiding the use of the word "stringer" insofar as practical. Therefore, shell stiffeners previously identified as stringers should be redesignated as longitudinals.

5.5.1 Shell longitudinal denotation. On ship construction drawings, shell longitudinals are identified by numbers. The number "1" designates the first longitudinal outboard of the vertical keel. This is followed by the numbers "2", "3", "4", etc., to the uppermost part of the shell. Preceding the number is a letter, the letter "L" being used to indicate a longitudinal (e.g., L1, L2, …, LN).

5.5.1.1 Deck longitudinal denotation. Deck longitudinals are numbered L1, L2, etc., beginning with the first longitudinal off centerline either port or starboard and increasing towards the outboard edge of the deck.

5.5.1.2 Longitudinal denotation on contract drawings. Longitudinals on contract drawings and contract guidance drawings are not numbered.

5.6 Welding and mechanical fasteners.

5.6.1 Welded and mechanically fastened seams and butts. Welded and mechanically fastened seams and butts are indicated on the drawing by an appropriate symbol in accordance with figure 1. (When the type and size of a weld or mechanical fastener are a contract requirement, these details will be shown on guidance drawings.)

5.6.2 Mechanical fasteners type and size symbols. The types and symbols for heads and points of mechanical fasteners are listed in table VII.

TABLE VII. Types of mechanical fasteners.

Type of part	Navy symbol	DoD symbol
Heads		
Button head	B	(BTNHD)
Countersunk head	C	(CSKH)
Tap	T	(TAP)
Points		
Countersunk point	CK	(CTSKPT)
Button point	B	(BTNPT)
Cone point	CP	(CP)

5.6.2.1 Fastener denotation. On drawings, following the dimension indicating the mechanical fastener diameter, the type of mechanical fastener shall be designated by a fraction; the numerator designates the head and the denominator the point, thus:

B/CK (BTNHD/CTSKPT) - Button head and countersunk point

B/B (BTNHD/BTNPT) - Button head and button point

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5.6.2.2 Rivets. The symbol HSS shall be used to designate higher strength steel rivets, and the symbol AL shall be used to designate aluminum alloy rivets. This symbol shall precede the dimension for rivet diameter, as HSS 3/B/CK (3/BTNHD/CTSKPT). Ordinary strength steel rivets shall not have any distinguishing symbol.

5.6.2.3 Special bolt-type fasteners. There are no standard symbols for certain special fasteners. Bolt styles and head types vary depending on manufacturer.

5.6.2.4 Zigzag and chain fasteners. Zigzag fastening shall not have any distinguishing symbol. Chain fastening shall be represented by the symbol CH.

5.6.3 Welding joint type and size symbols. Welding symbols for drawings shall be in accordance with AWS A2.4 and ASME Y14.38.

5.6.4 Welding and mechanical fastener applicability. Type and size welding and mechanical fastener symbols are generally only applicable to ship construction drawings; for contract designs, they are covered by specification requirements.

5.7 Application examples of symbols on ship structural drawings.

5.7.1 Contract drawings and contract guidance drawings. Typical examples showing the use of symbols on contract drawings and contract guidance drawings are shown on figures 2 through 6.

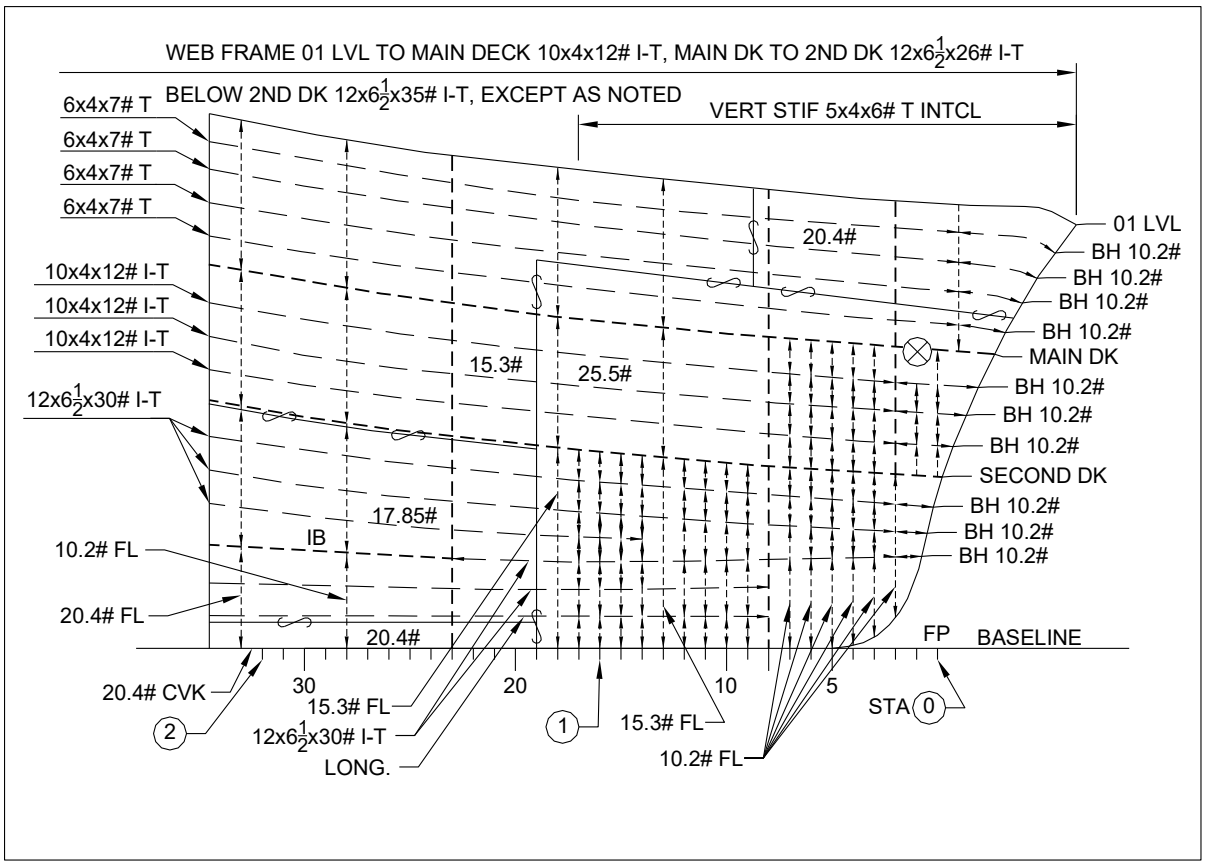


FIGURE 2. Contract drawing or contract guidance drawing – shell expansion.

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Check the source to verify that this is the current version before use.

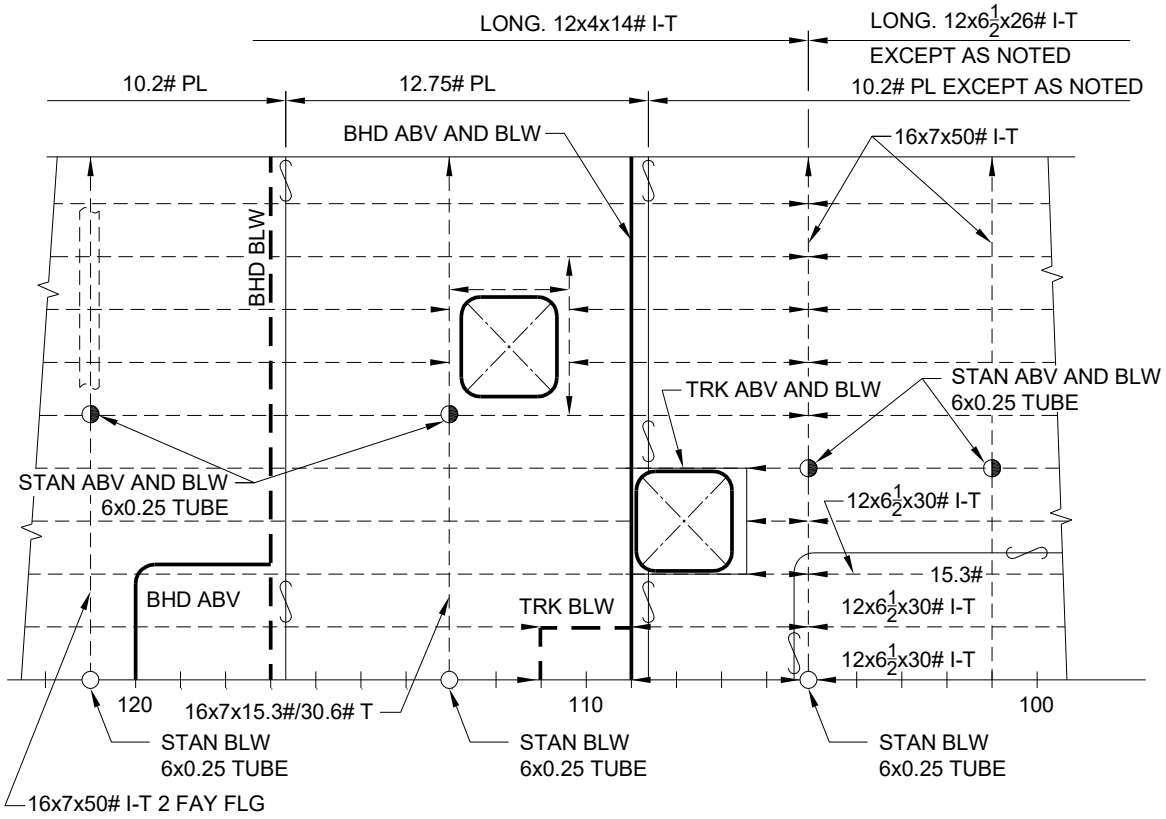


FIGURE 3. Contract drawing or contract guidance drawing – deck plan.

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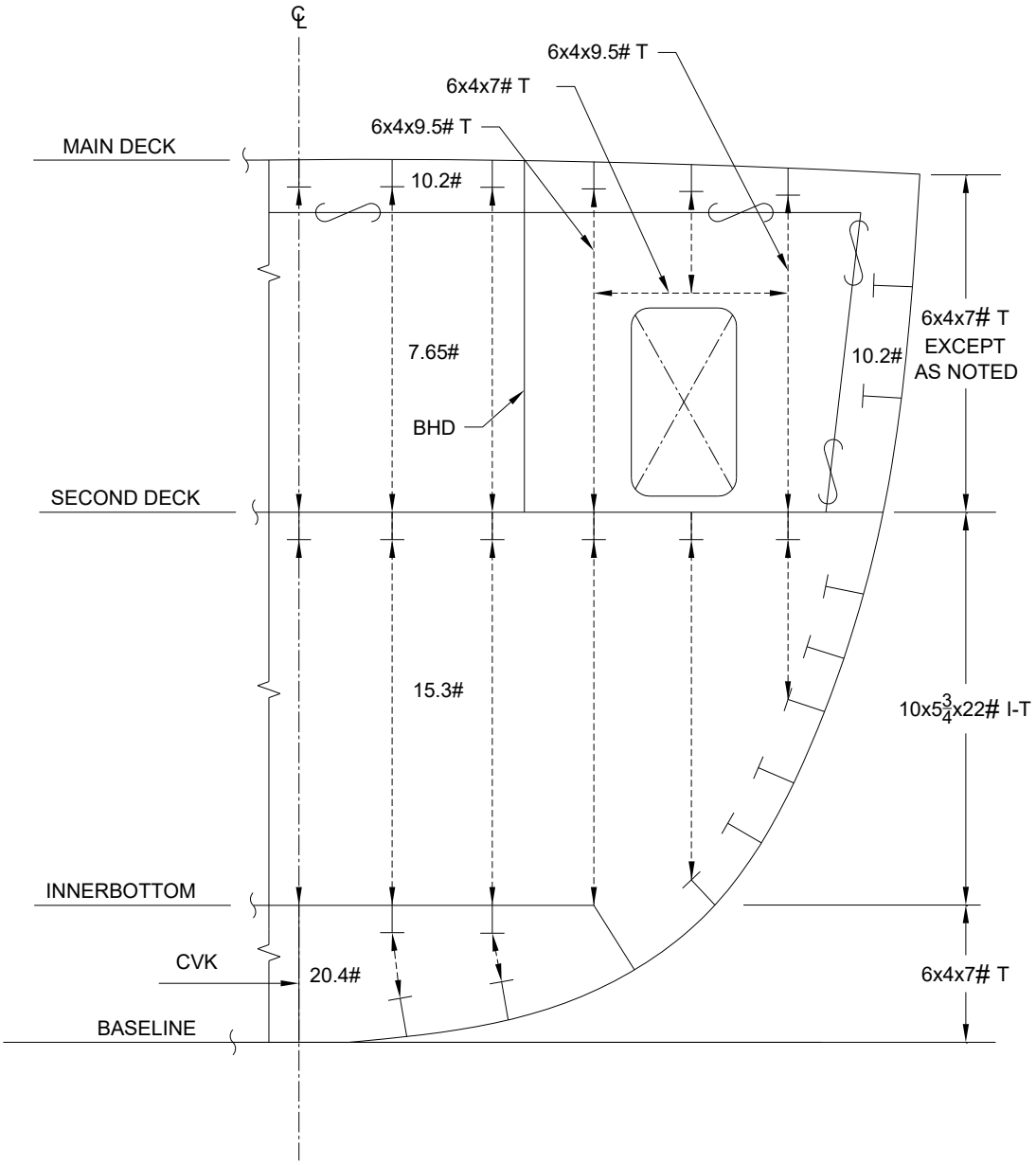


FIGURE 4. Contract drawing or contract guidance drawing – typical section, bulkhead.

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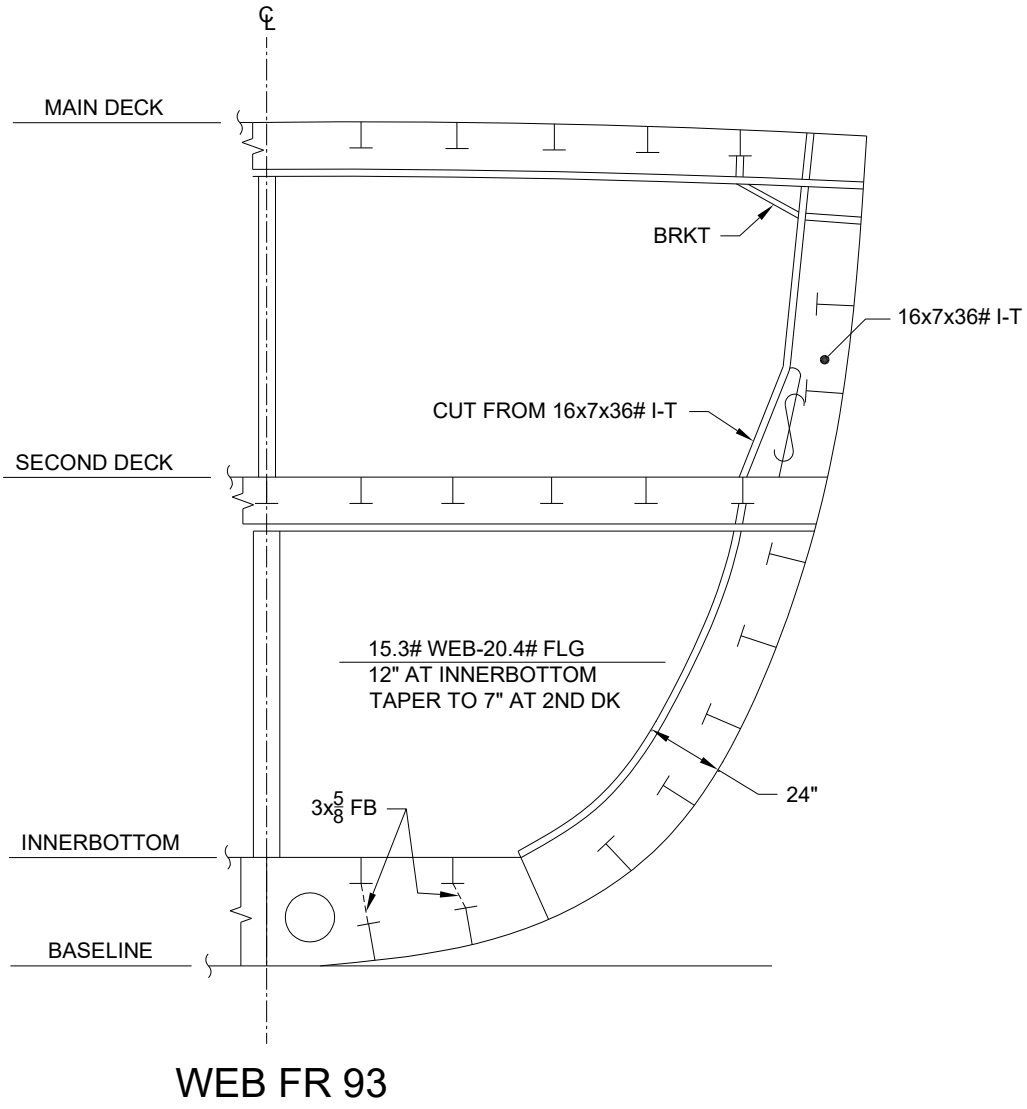


FIGURE 5. Contract drawing or contract guidance drawing – typical section, web frame.

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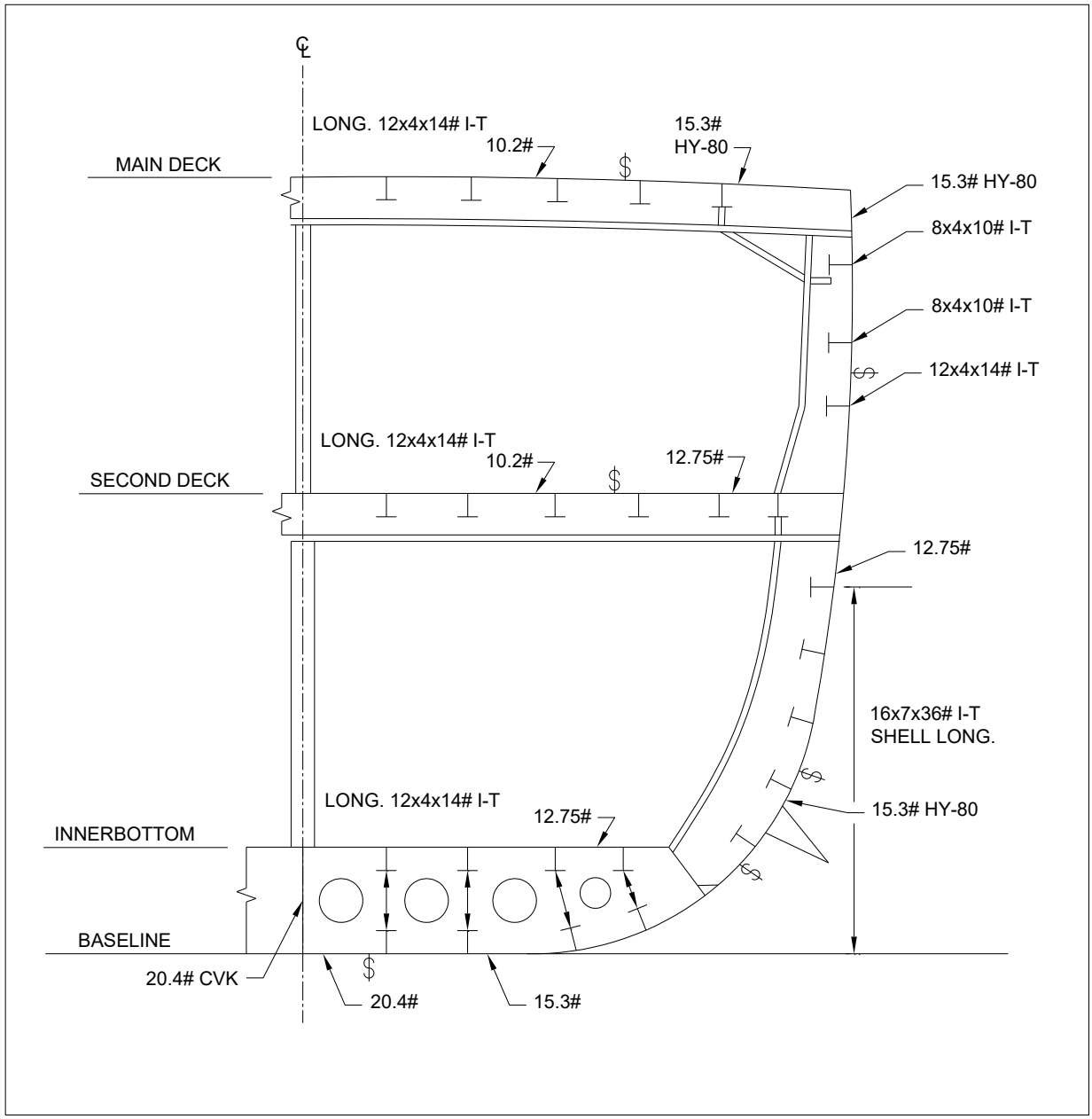


FIGURE 6. Contract drawing or contract guidance drawing – midship section.

5.7.1.1 Scantling identification on drawings. Note that in each example, scantlings are shown only for certain structures. For example, on the bulkhead drawing longitudinal stiffeners are not labeled because they are labeled on the shell expansion, deck drawing, and midship section. Table VIII indicates on which drawings scantlings designations shall be shown, to reduce repetitive identification.

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TABLE VIII. Scantling identification on drawings.

Scantling \ Drawing	Bulkhead and web frame drawings	Deck drawings	Shell expansion	Midship section
Longitudinal stiffeners	—	X	X	X
Transverse webs	X	X	X	—
Plating	X	X	X	X
Coamings	X	X	X	—
Vertical stiffeners	X	—	X	—
Vertical webs	X	—	—	—
Stanchions	—	X	—	X

5.7.1.2 Midship section drawings. The midship section drawing is a unique drawing. This drawing is sometimes designated a contract drawing, and only those scantlings that are desired as contractual items are labeled. In the example shown on figure 6, all structure contributing to the longitudinal strength of the hull is labeled. The transverse structure is considered contractor developed and therefore not labeled. This drawing will vary in content depending on the contractor scantlings invoked.

5.7.2 Ship construction drawings. Typical examples showing the use of symbols on ship construction drawings are shown on figures 7 and 8.

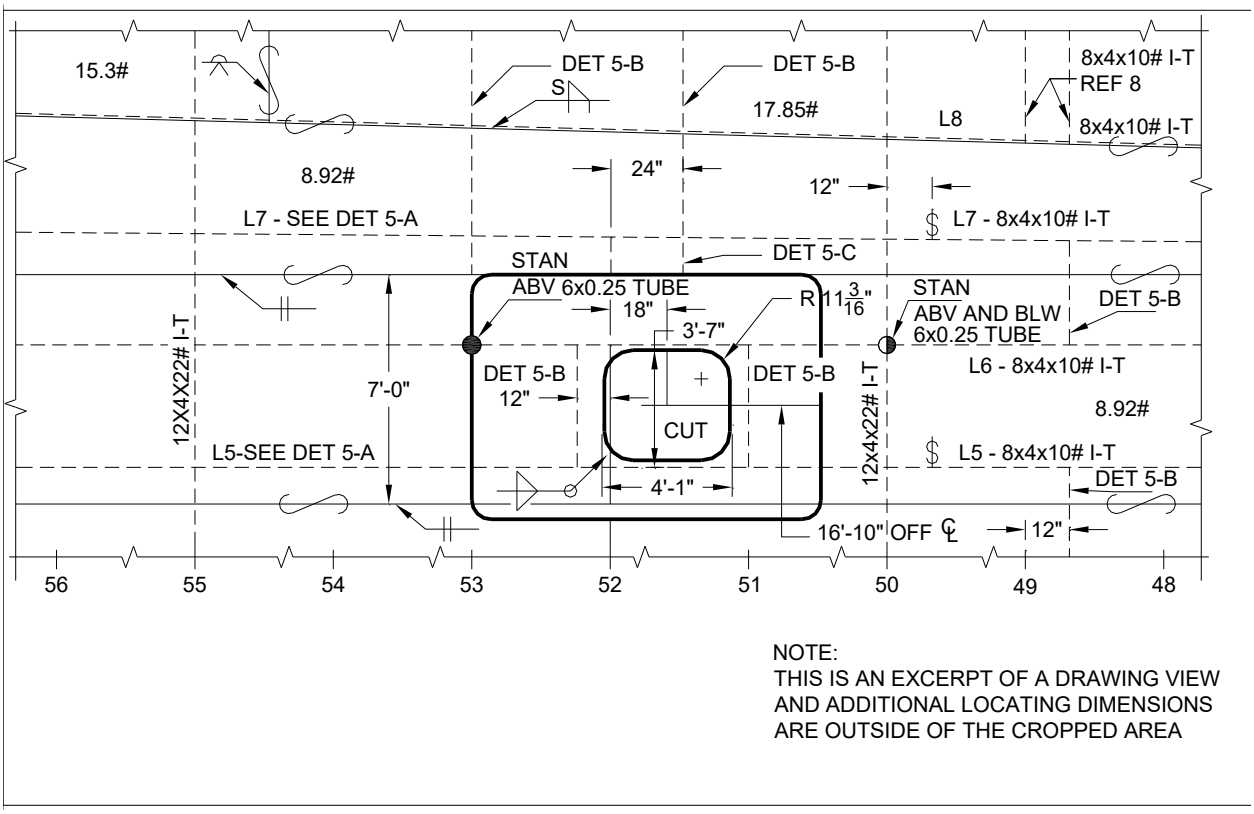


FIGURE 7. Ship construction drawing – deck drawing.

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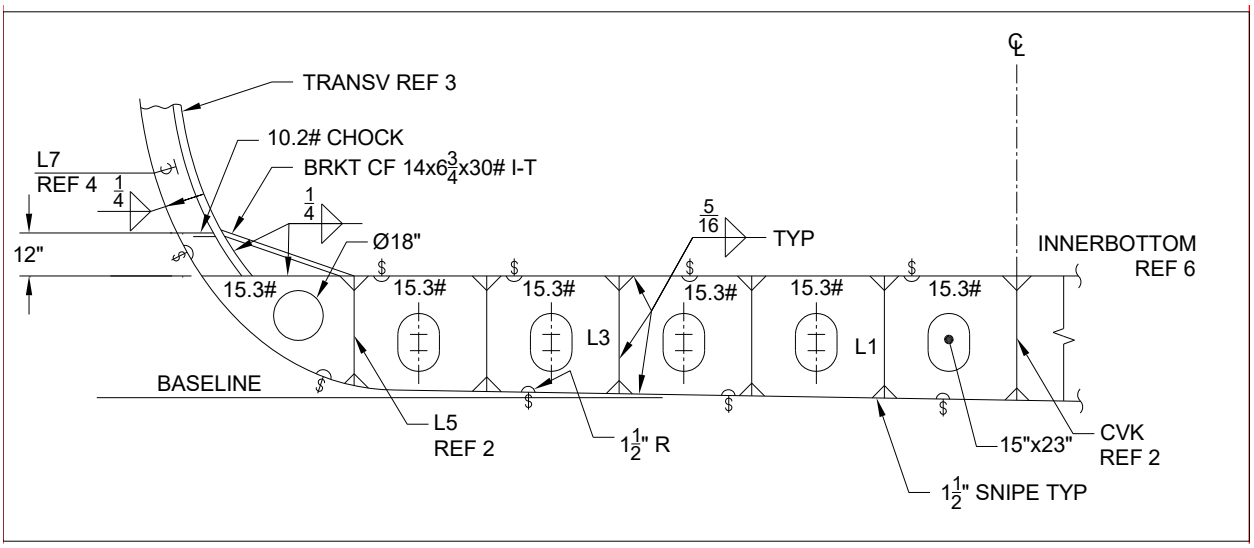


FIGURE 8. Ship construction drawing – nontight floor.

5.8 Metric conversion. Table IX shows metric symbols and conversion factors.

TABLE IX. Selected SI conversion factors.

Category	To convert from inch-pound units	To SI units 1/	Multiply by
Length	foot (ft) inch (in) in	meter (m) m millimeter (mm)	0.3048 2.540 10^{-2} 25.4
Area	ft^2 in^2	m^2 mm^2	9.290 10^{-2} 6.452 10^{2}
Force	kip pound-force (lbf)	newton (N) N	4.448 10^{3} 4.448
Mass	pound (lb) ton (long, 2,240 lb)	kilogram metric ton	0.454 1.016
Stress (force/area)	$kip/inch^2$ $lbf/inch^2$	pascal (Pa) Pa	6.895 10^{6} 6.895 10^{3}
SI makes extensive use of prefixes to form decimal multiples; it officially establishes 16 prefixes. The five prefixes most frequently used are as follows:			
mega	M	1,000,000	$=10^{6}$
kilo	k	1,000	$=10^{3}$
centi	c	0.01	$=10^{-2}$
milli	m	0.001	$=10^{-3}$
micro	μ	0.000001	$=10^{-6}$

NOTE:
1/ SI – International system of units.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This military standard is intended for use by Government engineers and management personnel for guidance in the development of structural drawings for U.S. Navy surface ships. This military standard provides guidance for the drawing development for a variety of structural plans, elevations, and sections for use in the construction of combatant, special auxiliary, and assault ships of the U.S. Navy. This military standard may be used as training reference material.

6.2 Subject term (key word) listing.

Structural drawings
Drawing abbreviations
Drawing designations

6.3 Background documents. The following publications form a part of the background of this document to the extent specified herein. The following references can be used for development of the structural drawing package:

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-22 - Welded Joint Design

MIL-STD-31000 - Technical Data Packages

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.24 - Types and Applications of Engineering Drawings

ASME Y14.34 - Associated Lists

ASME Y14.35 - Revision of Engineering Drawings and Associated Documents

ASME Y14.100 - Engineering Drawing Practices

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – SH
Air Force – 16

Preparing activity:
Navy – SH
(Project DRPR-2025-001)

Review activity:
Army – CR4

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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